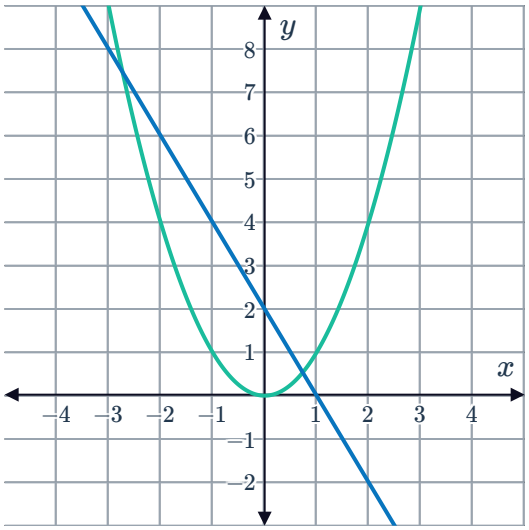
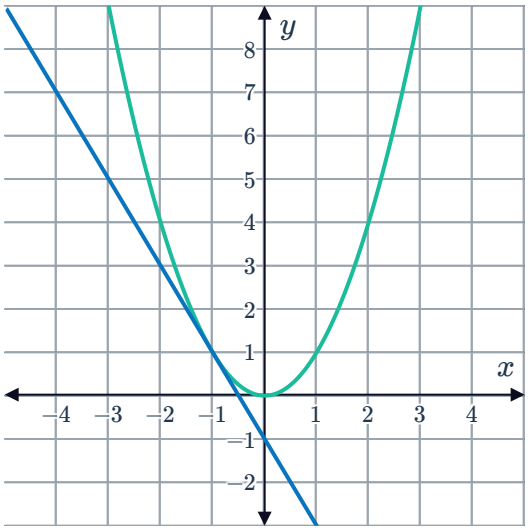
Points of intersection

1 State the number of points of intersection for each graph:

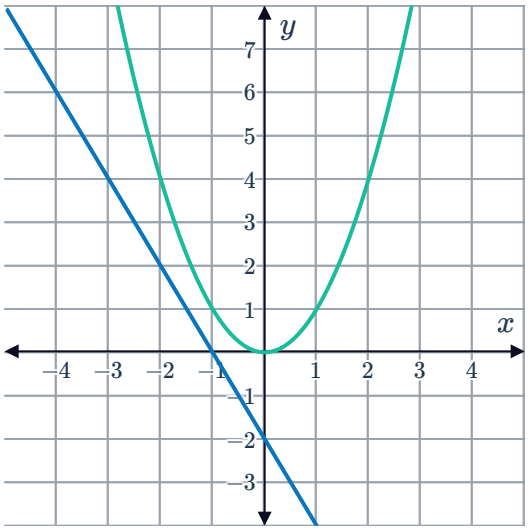
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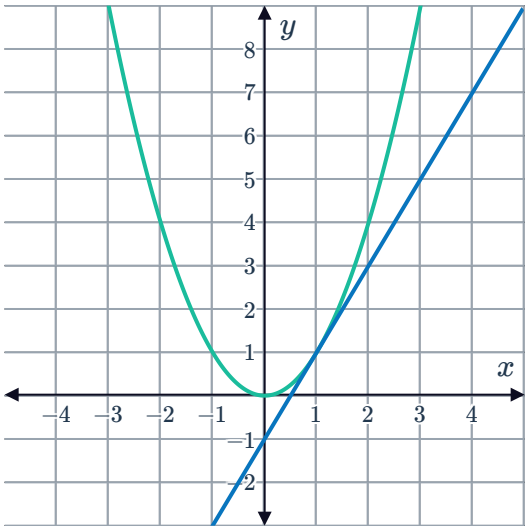
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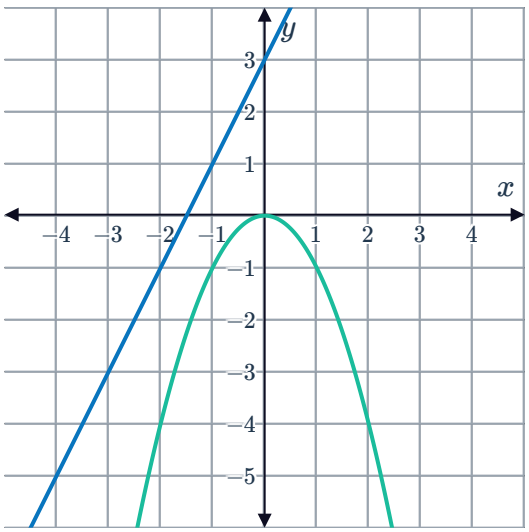
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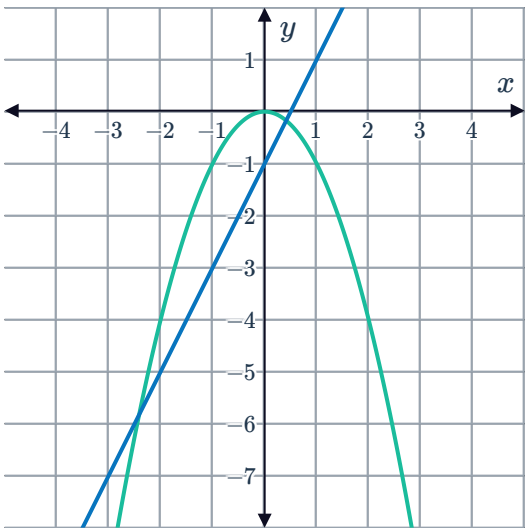
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e

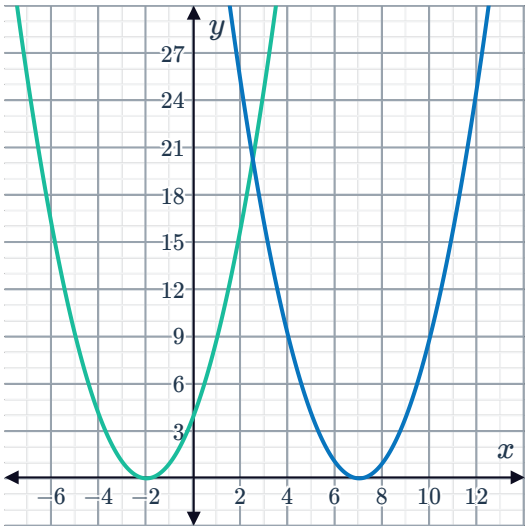


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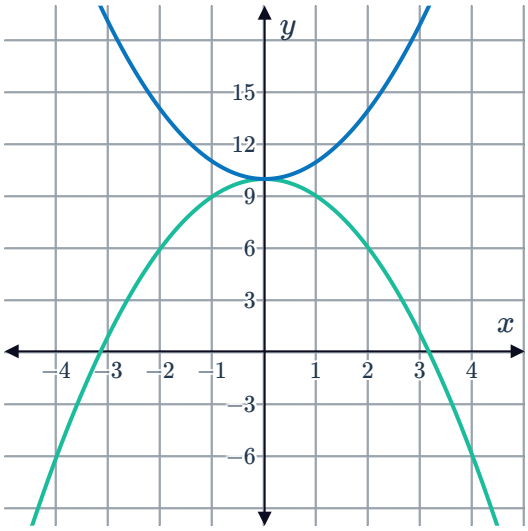


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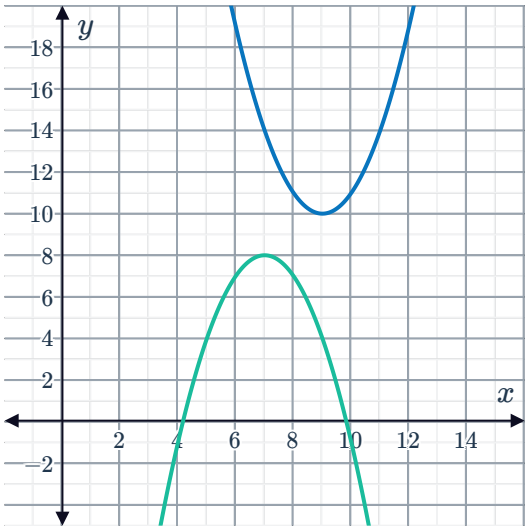
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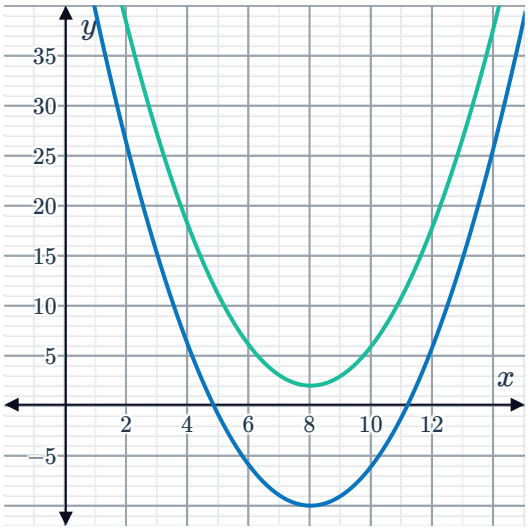
h



i



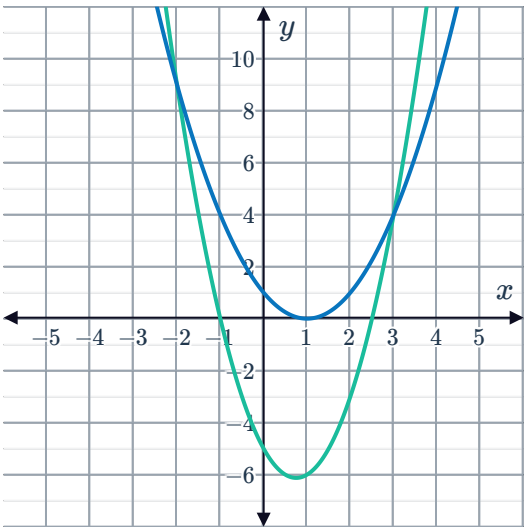
j



2 Consider the system of equations:

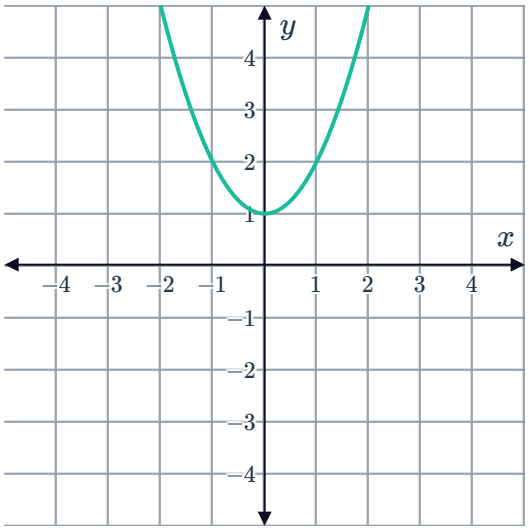
$$\begin{cases} y = 2x^2 - 3x - 5 \\ y = x^2 - 2x + 1 \end{cases}$$

What are the coordinates of the points of intersection?



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3 The function $y = x^2 + 1$ is shown on the graph: State the minimum value of k such that the line $y = k$ has at least one point of intersection with the parabola.



4 Consider the following system of equations: $\begin{cases} y = 4x^2 + 16 \\ y = 5x + 2 \end{cases}$

a Use the two equations to form a quadratic equation in terms of x only.

b Find the discriminant of the newly formed equation.

c Do the equations have any common coordinates? Explain your answer.

5 When solving $y = x^2 - 3x + 1$ and $y = x + 6$ simultaneously, one point of intersection is at $x = -1$. What is the y-coordinate at this point?

6 Find the coordinates of the points where the vertical line $x = -5$ intersects the curve $y = -2x^2 + x - 12$.

7 One point of intersection of the curves $y = x^2 - 2x + 30$ and $y = kx + 10$ occurs at $x = 4$. Find the value of k.

8 For each of the following pairs of equations:

i Graph the equations on the same coordinate plane.

ii State the coordinates of the points of intersection.

a $\begin{cases} y = x^2 + 4x + 1 \\ y = -x^2 + 7 \end{cases}$

b $\begin{cases} y = -3x + 4 \\ y = 8 - x^2 \end{cases}$

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9 Consider the following system of equations: $\begin{cases} y = -4x^2 + 10x \\ y = c \end{cases}$

a Find the x-intercepts of the quadratic equation.

b Find the axis of symmetry of the graph corresponding to the quadratic equation.

c Now, determine the coordinates of the vertex for the quadratic equation.

d Graph the quadratic equation on the coordinate plane.

e For what value of c will there be one solution to the system of equations?

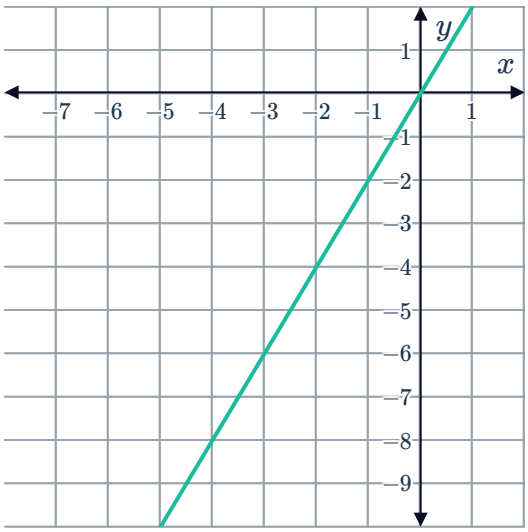
f For what values of c will there be two solutions to the system of equations?

10 Consider the equations $y = x^2 + 6x$ and $y = 2x$.

a Complete the table of values for the two equations.

x	-4	-3	-2
$y = x^2 + 6x$			
$y = 2x$			

b The line $y = 2x$ has been graphed. On the same coordinate plane, graph $y = x^2 + 6x$.



c Now state the coordinates of the points of intersection.

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11 Consider the equations $y = 4x - x^2$ and $y = -2x + 5$.

a Complete the table of values for the two equations.

x	1	2	3
$y = 4x - x^2$			
$y = -2x + 5$			

b Graph $y = 4x - x^2$ and $y = -2x + 5$ on the same coordinate plane.

c Now state the coordinates of the points of intersection.

12 Consider the parabola $y = x^2 - 3x - 3$ and the line $x = -2$.

a If there is only one point of intersection between the two graphs, does this mean $x = -2$ is a tangent to the parabola?

b State the coordinates of the point of intersection.

13 Vanessa is using algebra to find the points of intersection of a parabola and a straight line. She finds a quadratic equation in the form $ax^2 + bx + c = 0$, but when she tries to solve it, she finds that the equation has no solutions.
Graph an example of a parabola and a straight line that would satisfy this situation.

14 For each of the following pairs of equations:

i Sketch the graphs of the two equations on the same coordinate plane.

ii State the coordinates of the points of intersection.

a $\begin{cases} y = x^2 \\ y = 2x^2 - 1 \end{cases}$

b $\begin{cases} y = x^2 + 1 \\ y = 3 - x^2 \end{cases}$

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Quadratic systems of equations

15 For each of the following pairs of equations:

i Solve for the x-coordinates of the points of intersection.

ii State the coordinates of the points of intersection.

a $\begin{cases} y &= x^2 - 4 \\ y &= 4 - x^2 \end{cases}$

b $\begin{cases} y &= x^2 \\ y &= 4x^2 - 12 \end{cases}$

c $\begin{cases} y &= x^2 \\ y &= x^2 + 4x \end{cases}$

d $\begin{cases} y &= (x+2)^2 \\ y &= 34 - x^2 \end{cases}$

e $\begin{cases} y &= x^2 - 7x + 8 \\ y &= x^2 + 8x - 97 \end{cases}$

f $\begin{cases} y &= 2x^2 - 2x + 14 \\ y &= x^2 + 7x - 6 \end{cases}$

g $\begin{cases} y &= 11x^2 - 10x - 5 \\ y &= -5x^2 - 2x - 6 \end{cases}$

h $\begin{cases} y &= (x-8)^2 - 5 \\ y &= (x-9)^2 - 1 \end{cases}$

i $\begin{cases} y &= 2x^2 - 6x + 10 \\ y &= x^2 - 2x + 6 \end{cases}$

j $\begin{cases} y &= x^2 - 18 \\ y &= 2 - x^2 \end{cases}$

16 Find the points of intersection for each pair of equations:

a $\begin{cases} y &= x^2 \\ y &= -7x \end{cases}$

b $\begin{cases} y &= x^2 + 3x \\ y &= -x + 2 \end{cases}$

c $\begin{cases} y &= -2x - 1 \\ y &= x^2 - 8x + 8 \end{cases}$

d $\begin{cases} y &= 2x^2 + 14x \\ y &= 16x \end{cases}$

e $\begin{cases} y &= x^2 - 8x \\ y &= -7 \end{cases}$

f $\begin{cases} y &= (x-3)(x-2) \\ y &= -13x - 10 \end{cases}$

g $\begin{cases} y = -x^2 + 6 \\ x - y = 0 \end{cases}$

h $\begin{cases} y = (x-3)^2 \\ x - 2y = 3 \end{cases}$

i $\begin{cases} x^2 + y = 5 \\ 2y = 3x^2 - 10 \end{cases}$

17 Consider the following system of equations: $\begin{cases} y = -x^2 + x \\ y = -4 \end{cases}$

a Find the exact values of x of the points of intersection.

b Find the points of intersection in exact form.

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18 Find the points of intersection of the following two equations. Round your answers to two decimal places.

$$\begin{cases} y = -x^2 - 7x \\ x + y + 1 = 0 \end{cases}$$

19 Find the coordinates of the points of intersection for each of the following parabola and line pairs:

a The parabola $y = -x^2 - 3x + 1$ and the line $y = -3x - 3$.

b The parabola $y = x^2 + 7x$ and the line $y = -12$.

c The parabola $y = -x^2$ and the line $y = -2x$.

d The parabola $y = x(x - 4)$ and the line $y = -3x + 12$.

e The parabola $y = x^2 + 6x + 10$ and the line $y = -3x - 10$.

20 Consider the parabola $y = x^2 - 5$. Fiona discovered that the line with equation $y = \dfrac{4x - 21}{4}$ intersects the parabola at exactly one point.
Determine the equation of another straight line that has exactly one point of intersection with the parabola, and that has the same y-intercept as $y = \dfrac{4x - 21}{4}$.

21 Consider the following pair of parabolas:

$$\begin{cases} y = 2x^2 + 11x + 1 \\ y = x^2 + 3x - 3 \end{cases}$$

a Find the exact value of the x-coordinates of their points of intersection.

b Are the following statements true about these two parabolas?

i They do not intersect each other because one of the parabolas has greater concavity than the other.

ii They are tangents to each other at exactly one point.

iii They have two points in common.

22 The parabolas $y = c - ax^2$ and $y = x^2 + 2x + c$ intersect at the point $(2, 0)$.

a Solve for c.

b Solve for a.

c Graph the parabolas on the same coordinate plane.

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Quadratic equations from systems

23 The graphs of the two equations below are graphed and the x-coordinates of their points of intersection are found to be x_1 and x_2.

$$\begin{cases} y - 3x^2 - x - 7 = 0 \\ y = 3x - 2 \end{cases}$$

State the quadratic equation whose solutions are x_1 and x_2. Express the equation in the form $ax^2 + bx + c = 0$, where $a > 0$.

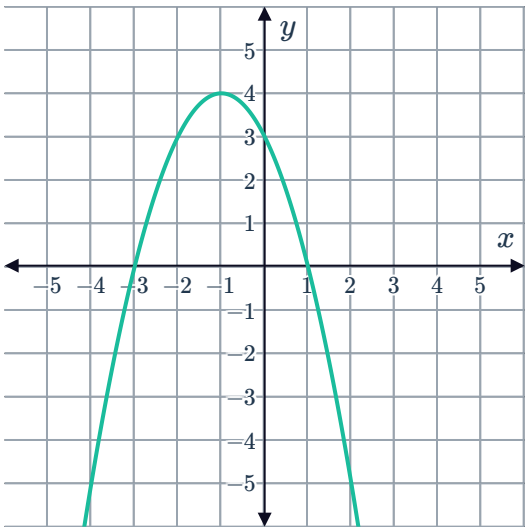
24 Consider the quadratic function $y = x^2 + 4$ and the linear function $x + y = 4$.

- **a** Sketch the graphs of $y = x^2 + 4$ and $x + y = 4$ on the same coordinate plane.
- **b** Determine the number of solutions to the equation $x^2 + 4 = -x + 4$.
- **c** Find the points of intersection of the two functions.

25 If you were given the graph of $y = x^2$, what suitable straight line would you need to graph on the same coordinate plane to solve the equation $x^2 - 2x - 6 = 0$?

26 The function $y = -x^2 - 2x + 3$ is shown on the graph:

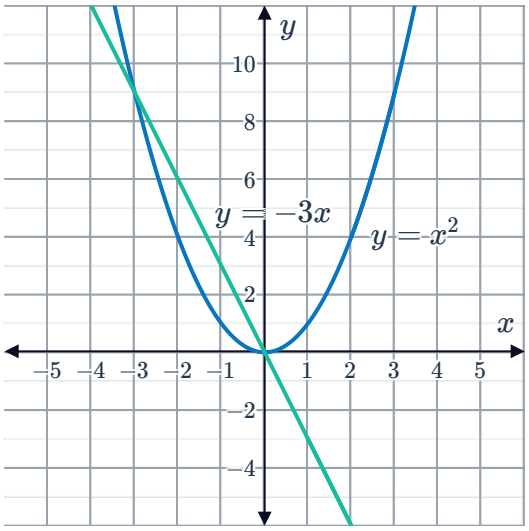
- **a** State the linear expression that must be added to $-x^2 + x + 2$ to form $-x^2 - 2x + 3$
- **b** Graph the straight line on the same coordinate plane.
- **c** Now determine the solutions to $-x^2 + x + 2 = 0$.



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27 The following functions are displayed on the given graph:

$$\begin{cases} y &= x^2 \\ y &= -3x \end{cases}$$



a How many points on the plane are common to both graphs?

b State the coordinates of the points of intersection.

c Now state the solutions to the equation $x^2 = -3x$.

28 Consider the functions $y = 9$ and $y = x^2$.

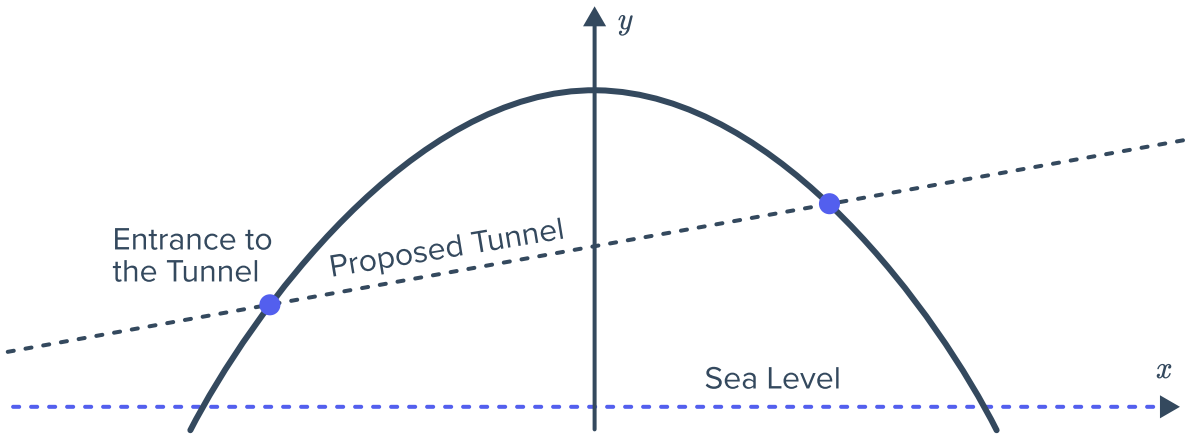
a Graph the functions on the same coordinate plane.

b State the coordinates of the points of intersection.

c Now state the solutions to the equation $x^2 = 9$.

Applications

29 The plan for a proposed tunnel through a mountain side is shown in the following diagram:



The height of the mountain above sea level is given by the equation $y = -\frac{x^2}{3} + 4$, and the height of the tunnel above sea level is given by the equation $y = \frac{x}{3} + 2$, where y represents the height above sea level in kilometers.
Solve for y, the heights above sea level at which cars will enter and exit the tunnel.

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30 In business, supply and demand functions are ways of expressing the supply or demand of a commodity as a function of its unit price. Market equilibrium is the term that describes when the supply is equal to the demand. Consider the supply and demand functions for a new power drill below, where y is in dollars and x is in thousands of units supplied.

Supply function	$y = 4x^2$
Demand function	$y = -6x + 5x^2$

a Solve for the value of x, where $x > 0$.

b Find the corresponding value of y.

c Interpret these values of x and y in context.

31 Sketch the graph of:

$$h(x) = \begin{cases} 3x - 2, & x < 0 \\ x^2 - 3x - 4, & 0 \leq x < 4 \end{cases}$$

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